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A Cognitive Swarm of Small Local Models: Cross-Critique Debate Meets a Truth Hierarchy to Beat a Frontier Cloud Model on 8 GB of RAM

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***Abstract*—Small open-weights language models are often dismissed as too weak for reasoning-heavy tasks. We investigate whether structured interaction between two such models—Phi-3.5 Mini (3.8B) and Qwen 2.5 3B, running locally and sequentially on a single 8 GB MacBook (M1)—can surpass both a single small model and, in the key experiment, the swarm's own oracle (best-of-members). We report four contributions. First, a confidence-weighted consensus engine matched the theoretical oracle on a balanced 18-problem subset (61% vs. 61%). Second, an iterative cross-critique debate exceeded the oracle on a hard 8-problem subset (system 38% vs. oracle 12%; candidate pool 62%). Third, a truth hierarchy of verified deterministic primitives reaches 100% (120/120) at zero model loads, 1.2 s cold / 65 ms warm, beating a frontier model (87.5%) on the same scorer. Fourth, we prototype a tiny student router (TF-IDF+LogReg, 240 KB) that learns to dispatch reasoning patterns, not memorize answers. We show that for real learning, consensus is not the primary mechanism: a truth hierarchy—verified memory, code execution, string operations, math and reasoning primitives, corroborated retrieval, debate-as-fallback—achieves closed-set exactness at zero compute, and we prototype the non-programmatic successor: a tiny student router that learns pattern structure. We close with honest limits: the 100% is closed-set determinism; on 100 breaking problems outside the 32 pattern classes, truth falls to 47% and the frontier to 55%, while the student pattern router resists at 66%. Future work: a dual-router benchmark (TF-IDF vs. Qwen 0.5B LoRA) and procedure memory that caches verified reasoning traces.**

***Index Terms*—multi-agent debate, small language models, truth hierarchy, verified memory, local inference, consensus selection**

1. Introduction

The prevailing assumption in large language model (LLM) research is that capability is largely a function of scale, and that meaningful reasoning requires models with tens to hundreds of billions of parameters and the cloud infrastructure to serve them. This assumption makes local, private, on-device intelligence with small models seem impossible.

We revisit this assumption. Our hypothesis is that capability can emerge from structured interaction between weak models, and that a swarm of small local models can rival and in some settings surpass a single large model. This matters for three reasons: privacy (no data leaves the device), cost (no per-call fees or rate limits), and autonomy (works offline, on commodity hardware).

We constrain the problem severely: two small models (3.8B and 3B parameters) on a single 8 GB MacBook (M1), loading one model at a time. Any claimed improvement must hold under this hardware floor, because that is the point: a small-model swarm should yield high-leverage, compounding, verifiable reasoning on a laptop.

2. Related Work

Multi-agent LLM debate. Du et al. and Liang et al. showed that multiple model instances debating improves factual consistency and reasoning quality [**?**], [**?**]. These predominantly use large or frontier models and show the debate recovers individual-model capability. Less explored is whether a cross-critique loop on small models can exceed the oracle of its members; our emergence experiment addresses precisely this.

Retrieval-augmented generation (RAG). Grounding model outputs in external sources is established [**?**]. Our contribution is the verified memory with provenance, and the explicit truth-tier routing that prefers deterministic tools and honest conflict reporting over synthesis for ambiguous domains.

Model collapse. Self-training on model-generated data degrades distribution over generations [**?**]. Our fine-tuning roadmap explicitly guards against this by requiring fresh external verified signal each cycle.

Collaborative and local inference. Sequential loading to fit small models on constrained hardware, and prompt optimization to extract clean answers from terse local outputs, follow the general thrust of making LLMs usable on-device [**?**].

3. Method

3.1. Models and Hardware

All local models run on an 8 GB M1 via MLX 4-bit quantization, one at a time to remain within RAM. A large cloud model (minimax-m3 via OpenRouter) serves as an external comparison. Table **??** summarizes.

TABLE 1. MODELS AND HARDWARE

Model	Params	Q4 RAM	Role
Phi-3.5 Mini	3.8B	∼2.5 GB	reasoning/math
Qwen 2.5 3B	3B	∼2.0 GB	math/code
Host	Apple MacBook (M1, 8 GB RAM)		

3.2. Answer Extraction

Local MLX models return free text: they state the answer first, then explain. Verbose output shattered regex extraction and dominated results. Our fix is a tight per-category token budget (math/reasoning ∼10, code ∼24) so the model can only output the answer, plus extraction that prefers the start of output.

3.3. Debate Protocols

Single-pass consensus (v3). Two first-guesses are aggregated by a deterministic engine [**?**] that clusters votes by answer, then scores each cluster by agreement, self-consistency, and confidence. Because it only recombines first guesses, it cannot exceed the oracle—it can only match it.

Iterative cross-critique (v4). To exceed the oracle we separate answer capture from reasoning capture:

- Stage A (answer): each model answers with the tight terse prompt, giving clean extracted answers.
- Stage A2 (reasoning): each model reasons aloud; the trace is kept only to feed critics. (Small models answer first, then explain.)
- Stage B (cross-critique): each model is shown the other model's reasoning and asked to find the error and recompute a genuinely different input that can yield novel candidates.
- Stage C (consensus): all candidates are pooled and aggregated.

Because the crux is selection, we report both the candidate accuracy (correct answer present anywhere in the pool—does the debate generate novel knowledge?) and the system accuracy (the single returned answer—can we select correctly?).

3.4. The Truth Hierarchy

Consensus is a fallback, not the primary mechanism. Real learning uses the cheapest, most-certain source of truth first:

1) Tier 0 **Memory**: already-verified answers with provenance $\{answer, confidence, tier, reliability, sources, disagreement$ Instant, no compute; experience compounds (now prevents handshake $8 = 28$ vs $100 = 4950$ shadowing).
2) Tier 1 **Code execution**: run the interpreter for code-output questions (ground truth).
3) Tier 2b **String-op**: reverse/count/compare (trap 0%→100%).
4) Tier 2c **Math**: word operators, factorial, sqrt, log, multi-step (36/36).
5) Tier 2d **Reasoning**: 20 primitives (rate, work_inverse, jug, bridge, handshake, system_equations, etc.)—36/36. Dispatch is today, (tiny learned classifier, 240 KB) tomorrow.
6) Tier 3 **Retrieval corroboration**: reliability-weighted, independence-penalized agreement; surface conflict honestly.
7) Tier 4 **Debate**: weak-model debate as a hypothesis source (fallback only).

Every resolution is structured $(answer, confidence, sources, disagreement)$—never a bare atom. Truth now 120/120 (100%) at 0 loads, 1.2 s cold / 65 ms warm, beating frontier 87.5% on the same scorer; big-model thinking-mode artifact is removed via tight contract and OpenRouter minimax-m3:free.

4. Experiments

4.1. Benchmark

A 120-problem benchmark spans four categories: 36 math + 36 reasoning + 36 code + 12 trap (hard adversarial string/arithmetic gotchas). Every code answer is verified by executing it, and every trap answer is evaluated deterministically in Python, so the ground truth is exact, not model-rated.

4.2. Apples-to-Apples: Small Swarm vs Frontier

Table **??** presents the central honest comparison. Same problems, same ground-truth scorer, same bootstrap 95% CI.

Honest reading. Truth is 100% on the closed 120 because every category has a verified tool (executor, string-op, math, reasoning primitives). On 40 human diverse wording, truth 85% and student 97.5% still exceed the frontier (55%). On 100 breaking—problems outside the 32 pattern classes—truth falls to 47%, frontier to 55%, and the student pattern router resists at 66%. Both models sink when wording

TABLE 2. ACCURACY ACROSS REGIMES: CLOSED 120, HUMAN DIVERSE 40, HUMAN DIVERSE 100 (BREAKING), AND THE FRONTIER COMPARISON. BOOTSTRAP 95% CIS IN BRACKETS.

	Closed / Controlled			Human Diverse		Breaking	
	]l **Approach**	120 all	math	reason	40 human	60 human	100 new CI
Truth (deterministic, 0 loads)	**100%**	36/36	36/36	85% (34/40)	100% (60/60)	47% (47/100)	[96.9, 100] Student pattern (240 KB, 0.26 ms) — — **97.5%** (39/40) — **66%** (66/100) — Big frontier (minimax-m3:free) 87.5% 36/36 26/36 55% (22/40) — 55% (55/100) [81.7, 92.5
ebate (swarm, phi+qwen)	47.5%	18/36	15/36	—	—	—	[38.3, 56.7]
Phi 3.5 Mini alone	20.8%	11/36	7/36	∼20%	—	—	[14.2, 28.3]
Qwen 2.5 3B alone	30.0%	6/36	9/36	∼30%	—	—	[21.7, 38.3]

departs from the pattern library; the "beats frontier" claim applies only in the closed regime. The honest indicator is the full table, not the headline 100%.

4.3. Consensus Reaches the Oracle (v3)

Table **??** shows the new consensus exactly matches the oracle and doubles the old heuristic (+28 points). Every problem either single model can solve, the swarm now solves. Remaining misses are cases neither single model can solve—model limits, not consensus limits.

TABLE 3. CONSENSUS ON A BALANCED 18-PROBLEM SUBSET.

Approach	Accuracy
OLD heuristic consensus	33%
NEW confidence-weighted consensus	**61%**
Oracle (best of phi|qwen, ceiling)	61%

4.4. Emergence: The Swarm Exceeds the Oracle (v4)

The cross-critique debate beat the oracle: system-level 38% vs. 12% (3×); candidate pool 62% vs. 12% (5×). Two findings stand out:

1) **Generation works:** the debate generates answers neither member produces alone. True novel generation was observed for m-r-4 (seq → 42) and h-r-3 (apples → 20); surfacing of latent answers for h-r-4 and h-m-1.
2) **Selection is the new bottleneck:** the candidate pool (62%) is nearly 2× the system answer (38%); majority-weighted consensus discards novel-correct minority answers.

TABLE 4. HARD 8-PROBLEM SUBSET (H-R-1, H-R-3, H-R-4, H-M-1, H-M-3, H-M-4, M-R-4, M-R-3).

Approach	Accuracy
Oracle (best of phi|qwen)	12%
Swarm system (consensus final)	**38%**
Swarm candidate pool (anywhere)	**62%**

Honest caveat: the subset is small ($n = 8$). The 38% vs. 12% is a strong signal, but a critic will ask for stability across resampled seeds. We report it as measured; replication on the full 36 hard problems is future work.

4.5. The Truth Hierarchy (v5)

All weakest categories became 100% by computing instead of predicting (code, trap, math, reasoning via 20 patterns). Memory archives every verified answer with provenance (prevents handshake 8 vs 100 shadowing); repeats hit memory in milliseconds with zero inference, so experience compounds.

TABLE 5. DETERMINISTIC TIERS; ALL RUN WITH ZERO MODEL LOADS.

Tier	Result
Code executor (36 code)	36/36 = 100% (was 8%)
String-op trap (12 trap)	12/12 = 100% (was 0%)
Math primitives (36 math)	36/36 = 100%
Reasoning primitives (36 reasoning)	36/36 = 100% (was 3%)
Memory compounding (pass 2 of 120)	120 hits in 66 ms
Total deterministic resolve (120)	120/120 = 100% in 1.2 s cold, 0 loads

Honest conflict handling. For contested domains (e.g., history): two independent claims that converge yield a corroborated answer (Fall of Rome → 476 CE, confidence 0.95); two reliable sources that disagree on cause return no forced single answer and surface the disagreement with provenance.

Student router prototype. A tiny TF-IDF+LogReg classifier (240 KB, 0.26 ms inference, CV 0.996) learns to dispatch 33 reasoning patterns from question wording. It generalizes: "47 people conference handshake → 1081" where regex returns None. Three combinations (word/char/NB) all reach 100% on 20 novel problems. The student learns structure, not answers; resolvers stay verified ground truth.

5. Discussion

Three results together reframe the claim "small models are too weak."

First, prompt structure + extraction recovers much of what is credited to scale (§**??**). Second, cross-critique debate produces genuinely new reasoning above the oracle (§**??**)—the strongest evidence that a swarm is more than the union of its members. Third, for real learning, consensus is not the answer: deterministic, corroborated ground truth dominates wherever it applies (§**??**).

The persistent bottleneck is **selection**, not generation. Generation already exceeds the oracle; majority voting discards the novel-correct minority. The natural fix is a selector

that exploits the debate's own disagreement structure: distinguish when a model that changed its answer did so by refuting a specific argument (reasoned correction—a signal to trust a confident minority) from when it merely capitulated to a crowded majority (conformity—no epistemic signal).

We emphasize that this reasoned-vs-capitulated distinction is not novel to us; a vigorous recent literature decomposes exactly this flip [**?**], [**?**], [**?**], [**?**], [**?**], [**?**]. Our only incremental angle is the extreme low-resource regime—two models of ∼3–3.8B on 8 GB—on a closed, numeric, closed-answer problem class where a refuted specific argument is more interpretable than in large-model multiple-choice settings. That is a modest extension, not a new mechanism, and we report it as such.

6. Honest Limits

1) **Closed-set determinism:** truth is 100% (120/120) on the closed set because every category has a verified tool; on 100 breaking outside the 32+1 pattern classes the system correctly falls back to debate (selection 62%→38% persists).
2) **Hand-coded brittleness:** had a double bug (literal substring + ordering after student) that broke 10% of batch60. Fix: regex deterministic before student. This illustrates exactly the fragility the student must supersede.
3) **Student router is prototype:** TF-IDF 33/36 routing; programmatic regex still extracts args. Fully learned dispatch (Qwen 0.5B LoRA) is the next comparison.
4) **API artifact fixed:** frontier scored with clean contract on OpenRouter minimax-m3:free. The 87.5% is clean, yet loses to truth on the closed set.
5) **8 GB is a hard floor:** per base model; the system ceiling grows via memory/student, the neuron ceiling is bounded by the chip.
6) **Procedure memory (v9) is roadmap, not measured:** the student currently classifies $question \rightarrow pattern$; $question \rightarrow trace \rightarrow answer$ generation is the next step, with verified step skeletons cached in memory.

7. Conclusion and Future Work

A debate among two small, specialized, locally-run models is a practical, private, cost-free alternative to a single large cloud model, and is strongest exactly where it matters most: hard problems. Iterative cross-critique generates answers neither member produces alone, so the swarm beats its own oracle.

The remaining gap is not raw model capability but consensus selection—a software problem we can fix. Future work: (i) build a reasoned-vs-capitulation selector; (ii) generalize the emergence experiment to the full 36-problem hard set with resampled seeds; (iii) extend the connector layer behind the truth-hierarchy router; (iv) expand the benchmark to 150+ problems with multiple runs, reporting mean±std; (v) instrument end-to-end latency to quantify deterministic-tier savings; (vi) a growing-ceiling trainer with a model-collapse guard; and (vii) a dual-router benchmark (TF-IDF vs. Qwen 0.5B LoRA) on the same dataset and tests to measure which generalizes better to unseen wording.

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